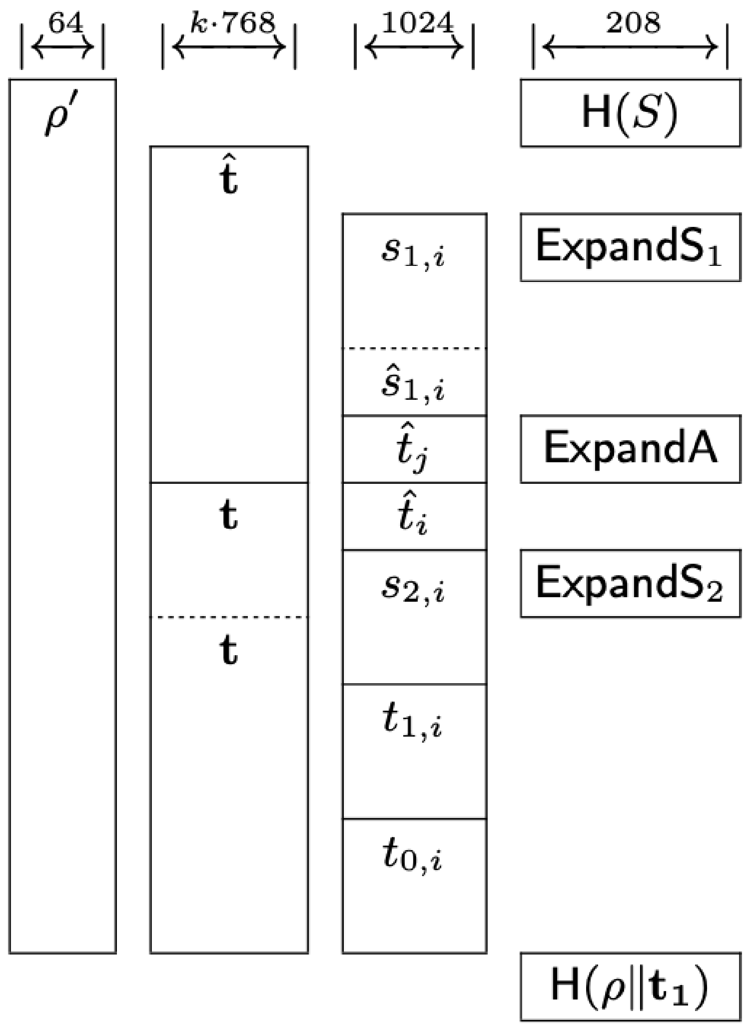
AI+Github+overleaf 그림 작성

https://youtu.be/qe5EeXOsx-A

정보컴퓨터공학과 송경주

Latex Figure 그리기



$\rho, \rho', K := \mathsf{H}(\zeta) \triangleright$ write ρ, K to output

$\hat{\mathbf{t}} := 0$

$0 \leq i < \ell$

$s_{1,i} = \mathsf{ExpandS}_1(\rho', i)$

write $s_{1,i}$ **to output**

$\hat{s}_{1,i} := \mathsf{NTT}(s_{1,i})$

$\hat{t}_j := \hat{t}_j + \hat{A}_{j,i} \circ \hat{s}_{1,i}$ **for** $0 \leq j < k$

$0 \leq i < k$

$t_i := \mathsf{NTT}^{-1}(\hat{t}_i)$

$s_{2,i} := \mathsf{ExpandS}_2(\rho', i)$

$t_i := t_i + s_{2,i}$

$(t_{1,i}, \_) := \mathsf{Power2Round}_q(t_i, d)$

write $t_{1,i}$ **to output**

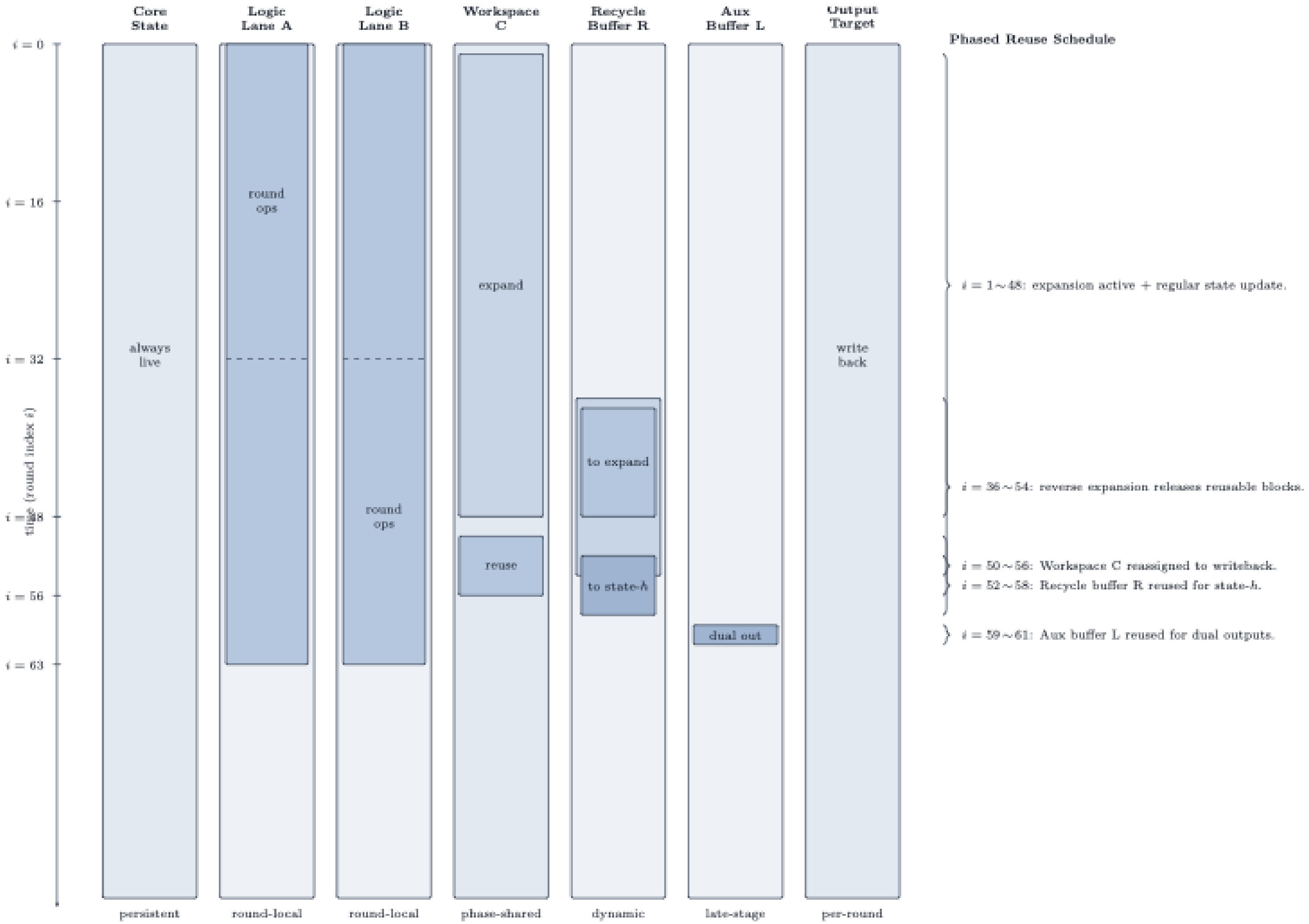
$(\_, t_{0,i}) := \mathsf{Power2Round}_q(t_i, d)$

write $t_{0,i}$ **to output**

$tr := \mathsf{H}(\rho \| \mathbf{t}_1) \triangleright$ write tr to output

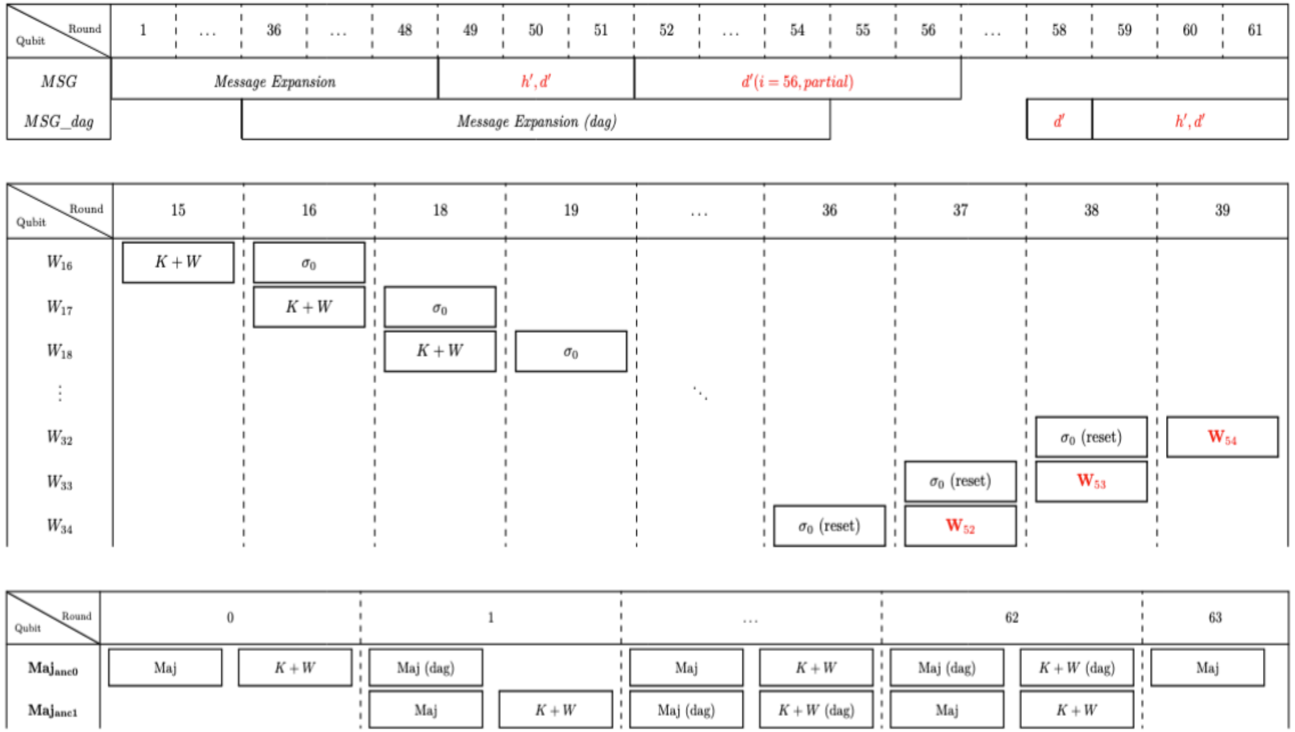
Latex Figure 그리기

- 스스로 코드 분석 후 그림 그려달라고 요청



Latex Figure 그리기

- 직접 PPT로 그리고, Latex로 변환해 달라고 함 (Gemini)




```
% 1번 테이블 전용 헤더 열 너비 고정 명령어
\newcommand{\fcolA}[1]{\makebox[1.15cm][c]{\makecell{#1}}}

% 다중 열을 가로지르는 박스의 위아래 여백을 2, 3번 테이블과 동일하게 맞추는 패딩 명령어
\newcommand{\padbox}[1]{\rule[-1.2ex]{0pt}{3.8ex}\makecell{#1}}

\begin{table}[hbt!]
\centering
\renewcommand{\arraystretch}{2.0} % 2, 3번 테이블과 동일한 세로 여백
\setlength{\arrayrulewidth}{0.6pt}
\resizebox{0.99\textwidth}{!}{
\begin{tabular}{|c|c:c:c:c:c:c:c:c:c:c:c:c:c:c:c:c|}
\hline
\multicolumn{1}{|c|}{\diagbox[width=2.5cm, height=1.1cm]{\scriptsize Qubit}{\scriptsize
Round}} &
\fcolA{1} & \fcolA{\dots} & \fcolA{36} & \fcolA{\dots} & \fcolA{48} &
\fcolA{49} & \fcolA{50} & \fcolA{51} &
\fcolA{52} & \fcolA{\dots} & \fcolA{54} & \fcolA{55} & \fcolA{56} &
\fcolA{\dots} &
\fcolA{58} & \fcolA{59} & \fcolA{60} & \fcolA{61} \\
\hline
$MSG$ &
\multicolumn{5}{|c|}{\padbox{\textit{Message Expansion}}} &
\multicolumn{3}{|c|}{\padbox{\textcolor{red}{$h', d'$}}} &
\multicolumn{5}{|c|}{\padbox{\textcolor{red}{$d' (i = 56, partial)$}}} &
\multicolumn{5}{c|}{} \\
```

Latex Figure 그리기

- 기존 그림은 Round 스케일이 달라 각자 테이블로 그려지고 깔끔하지X

  →깔끔하게 한 테이블로 만들어달라고 부탁함

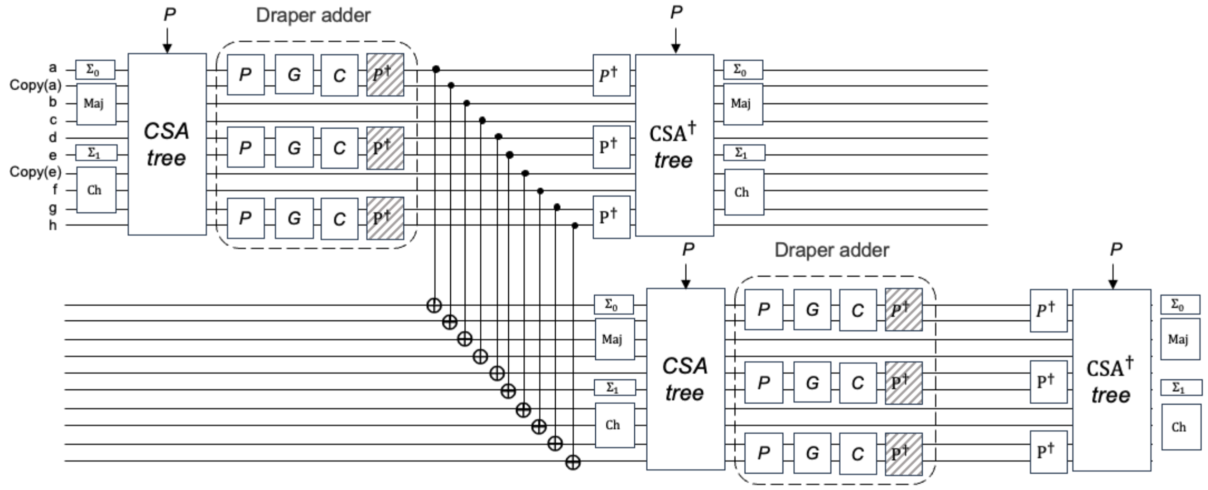
  문제점: 합치니까 사용 라운드에 대한 배치가 계속 틀려짐 → 따라서 잘 그려졌는지 파악해서 잘못된 부분 계속 말해주면서 수정 진행

- 그냥 AI로 그릴때는 깨지는 그림을 직접 확인해야했음
  - Github에 연결해서 쉽게 바로 확인하고 적용이 가능함
  - Github 연결로 오류도 바로 해결가능

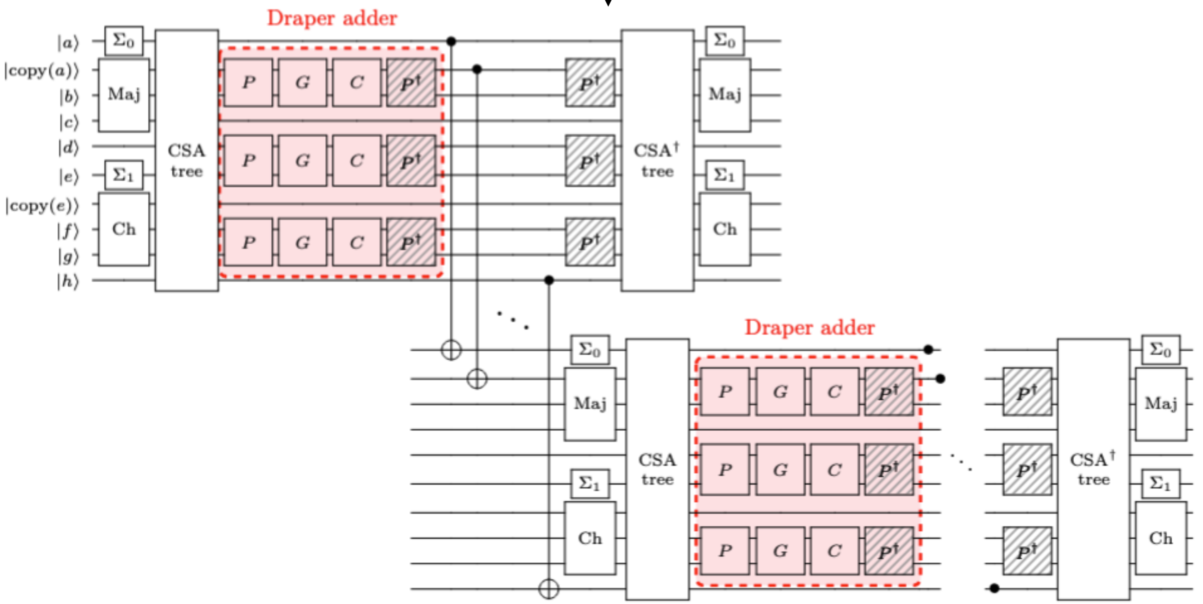
Rnd / Qubit	0	1	…	14	15	16–35	36–39	40–48	49–51	52–54	55	56	57	58	59	60	61	62	63
MSG		Message Expansion (W_0–W_{63})							h', d'	d'	d'	d' (pt)							
MSG†							Message Expansion†							d'	h', d'				
$\mathbf{W_{16}}$–$\mathbf{W_{34}}$					W_{16}: $K+W$	$K+W$, σ_0 (diagonal)	σ_0 reset, reuse $W_{52\text{–}54}$												
$\mathbf{Maj_{anc0}}$	Maj, $K+W$	Maj†	…	Maj, $K+W$	Maj†	…						Maj, $K+W$	Maj†	Maj, $K+W$	Maj†	Maj, $K+W$	Maj†	Maj†, $K+W^\dagger$	Maj
$\mathbf{Maj_{anc1}}$		Maj, $K+W$	…	Maj†	Maj, $K+W$	…						Maj†	Maj, $K+W$	Maj†	Maj, $K+W$	Maj†	Maj, $K+W$	Maj, $K+W$	

Latex Figure 그리기

- 직접 PPT로 그리고, Latex로 변환해 달라고 함 (Gemini)

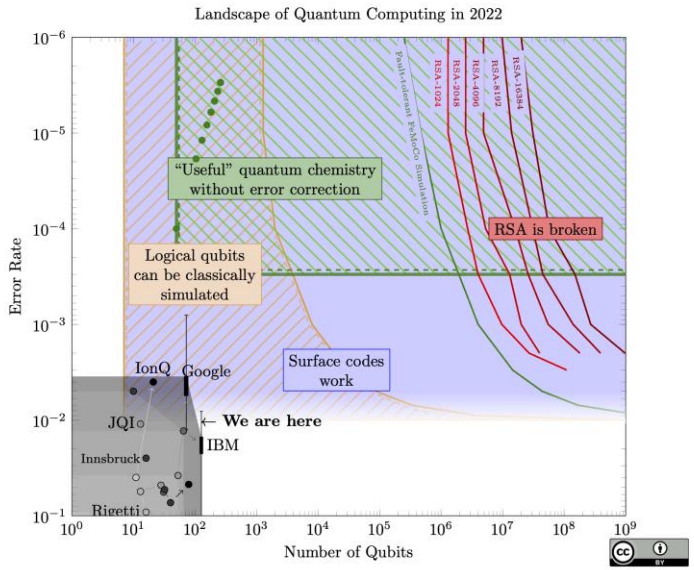



```
% 위쪽 Draper adder 박스 (빨간 점선, 파스텔 분홍색 배경, 둥근 모서리)
  \draw[red, dashed, very thick, fill=pastelpink, fill opacity=0.5, rounded corners=3pt]
      (pic cs:TLtop) rectangle (pic cs:BRtop);
  % 아래쪽 Draper adder 박스
  \draw[red, dashed, very thick, fill=pastelpink, fill opacity=0.5, rounded corners=3pt]
      (pic cs:TLbot) rectangle (pic cs:BRbot);
\end{tikzpicture}%
\Qcircuit @C=0.25em @R=0.15em {
% ========== R_i (Top Block) ==========
\lstick{\ys \scriptstyle |a\rangle} & \gate{\scriptstyle\Sigma_0} & \multigate{9}{\csalab} & \qw & \qw & \qw & \qw &
\ctrl{11} & \qw & \qw & \qw & \qw & \multigate{9}{\csadlab} & \gate{\scriptstyle\Sigma_0} & \qw & \push{\rule{1.5em}
{0em}} \\
\lstick{\ys \scriptstyle |\mathrm{copy}(a)\rangle} & \multigate{2}{\smaj} & \ghost{\csalab} & \multigate{1}{\sP} &
\multigate{1}{\sG} & \multigate{1}{\sC} & \multigate{1}{\sPd} & \qw & \ctrl{11} & \qw & \qw & \multigate{1}{\sPd} &
\ghost{\csadlab} & \multigate{2}{\smaj} & \qw \\
\lstick{\ys \scriptstyle |b\rangle} & \ghost{\smaj} & \ghost{\csalab} & \ghost{\sP} & \ghost{\sG} & \ghost{\sC} &
\ghost{\sPd} & \qw & \qw & \qw & \qw & \ghost{\sPd} & \ghost{\csadlab} & \ghost{\smaj} & \qw  \\
\lstick{\ys \scriptstyle |c\rangle} & \ghost{\smaj} & \ghost{\csalab} & \qw & \qw & \qw & \qw & \qw & \qw & \qw & \qw &
\qw & \ghost{\csadlab} & \ghost{\smaj} & \qw \\
\lstick{\ys \scriptstyle |d\rangle} & \qw & \ghost{\csalab} & \multigate{1}{\sP} & \multigate{1}{\sG} & \multigate{1}
{\sC} & \multigate{1}{\sPd} & \qw & \qw & \qw & \qw & \multigate{1}{\sPd} & \ghost{\csadlab} & \qw & \qw \\
\lstick{\ys \scriptstyle |e\rangle} & \gate{\scriptstyle\Sigma_1} & \ghost{\csalab} & \ghost{\sP} & \ghost{\sG} &
\ghost{\sC} & \ghost{\sPd} & \qw & \qw & \qw & \qw & \ghost{\sPd} & \ghost{\csadlab} & \gate{\scriptstyle\Sigma_1} & \qw
\\
\lstick{\ys \scriptstyle |\mathrm{copy}(e)\rangle} & \multigate{2}{\sch} & \ghost{\csalab} & \qw & \qw & \qw & \qw & \qw
& \qw & \qw & \qw & \qw & \ghost{\csadlab} & \multigate{2}{\sch} & \qw \\
\lstick{\ys \scriptstyle |f\rangle} & \ghost{\sch} & \ghost{\csalab} & \multigate{1}{\sP} & \multigate{1}{\sG} &
\multigate{1}{\sC} & \multigate{1}{\sPd} & \qw & \qw & \qw & \qw & \multigate{1}{\sPd} & \ghost{\csadlab} & \ghost{\sch}
& \qw \\
\lstick{\ys \scriptstyle |g\rangle} & \ghost{\sch} & \ghost{\csalab} & \ghost{\sP} & \ghost{\sG} & \ghost{\sC} &
\ghost{\sPd} & \qw & \qw & \qw & \qw & \ghost{\sPd} & \ghost{\csadlab} & \ghost{\sch} & \qw \\
\lstick{\ys \scriptstyle |h\rangle} & \qw & \ghost{\csalab} & \qw & \qw & \qw & \qw & \qw & \qw & \qw & \ctrl{11} & \qw &
\ghost{\csadlab} & \qw & \qw
% 위쪽 Draper 점선 박스
```

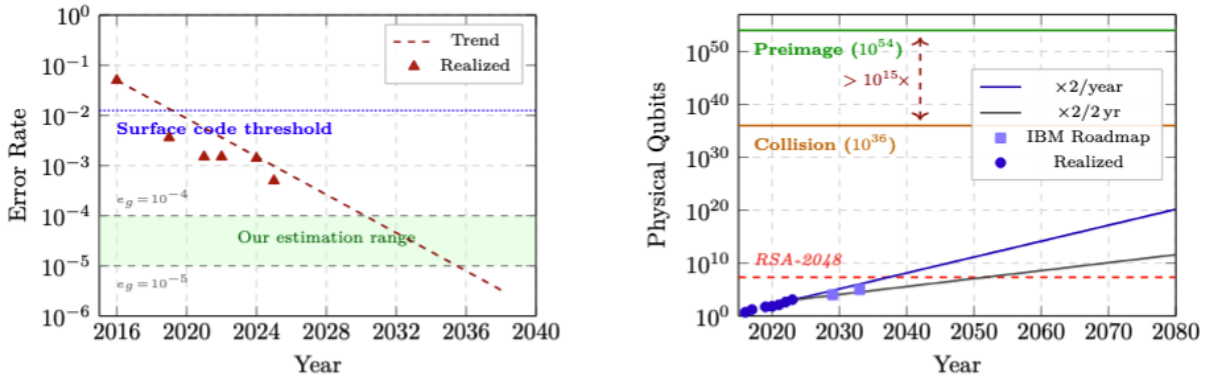


Latex Figure 그리기

- 자원 추정 결과로 이미지 작성
  - 자원추정 결과 + 그리고자 하는 이미지 제공 후, 원하는 이미지 요구
  - 큐비트 수, 오류율, 큐비트+오류(종합) 그래프를 나눠서 그려달라고 요청함

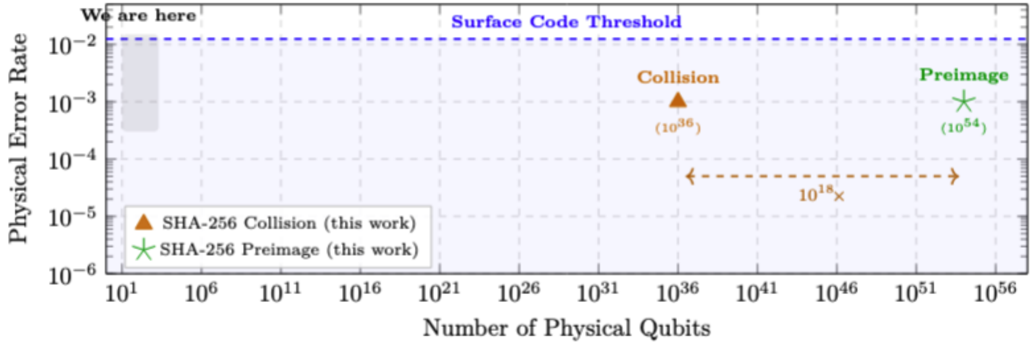


Attack	Hash	Distill. distances	L	Factories (Φ)	d_c	$\log_2 S$	Q_{phys} per instance	Q_{phys} total ($\log_2$)	Runtime (years)
Preimage (Grover)	SHA-256	$\{49, 19, 9\}$	3	616	49	76.5	8.16×10^7	$2^{179.3}$ ($\approx 10^{54.0}$)	3.1×10^{30}
	SHA-384	$\{69, 25, 11, 7\}$	4	1,150	69	141.1	1.07×10^9	$2^{312.1}$ ($\approx 10^{93.9}$)	1.2×10^{50}
	SHA-512	$\{87, 33, 13, 7\}$	4	1,150	87	205.1	1.71×10^9	$2^{440.8}$ ($\approx 10^{132.7}$)	2.7×10^{69}
Collision (CNS)	SHA-256	$\{33, 15, 7\}$	3	616	33	26.67	3.71×10^7	$2^{121.2}$ ($\approx 10^{36.5}$)	2.1×10^{15}
	SHA-384	$\{45, 19, 9\}$	3	1,150	45	65.60	1.35×10^8	$2^{222.2}$ ($\approx 10^{66.9}$)	1.5×10^{27}
	SHA-512	$\{57, 23, 11\}$	3	1,150	57	104.00	2.16×10^8	$2^{321.0}$ ($\approx 10^{96.6}$)	6.7×10^{38}



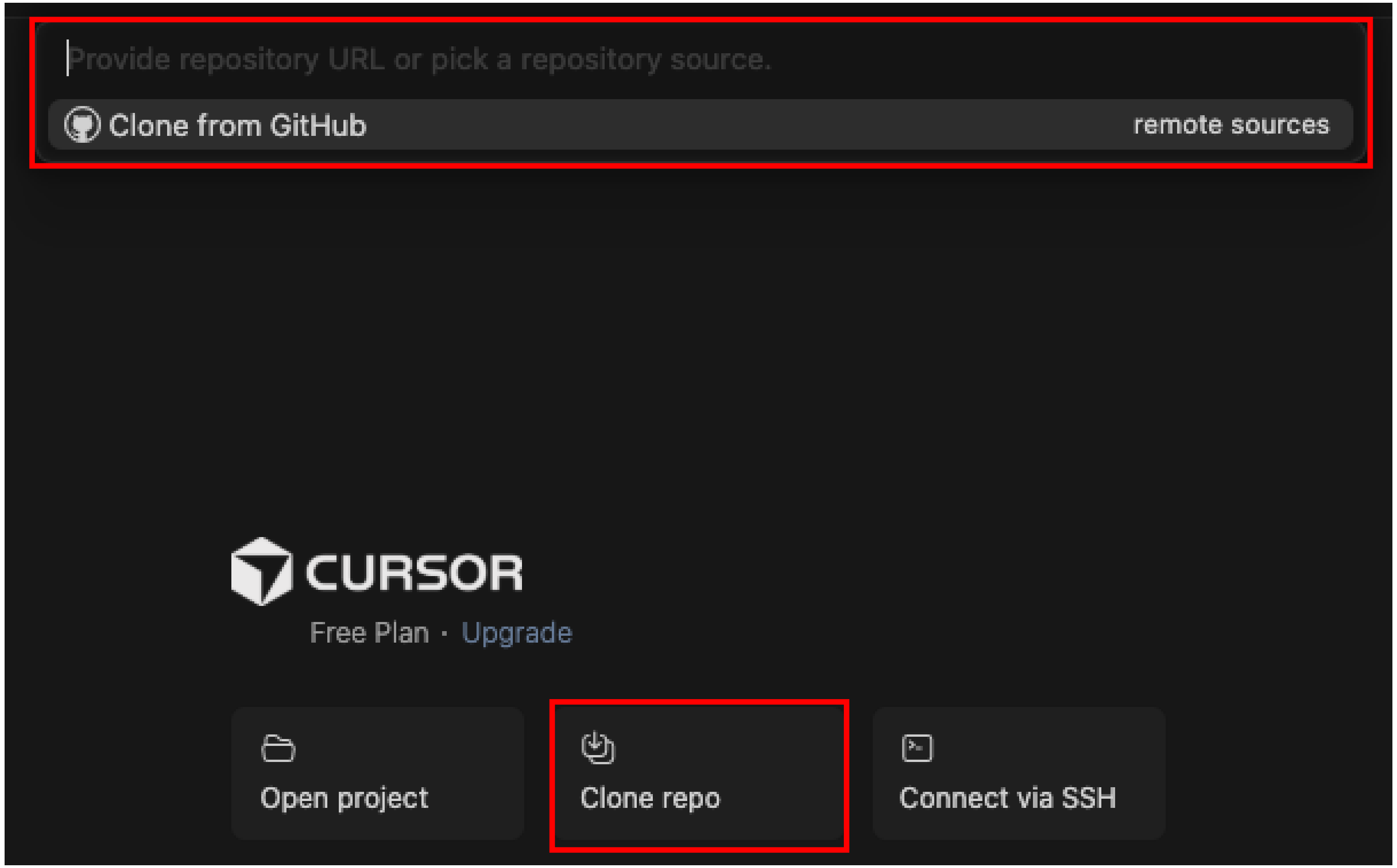
(a) Two-qubit gate error rate over time.

(b) Physical qubit count over time.



Latex Figure 그리기

- Cursor + Github 연결



Q & A